Electronic Service Quality and Customer Complaint Handling on Food Delivery Platforms: A Case Study of Nasi Gerilya on GrabFood

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ABSTRACT

Online food delivery platforms require small restaurants to manage service quality and customer satisfaction effectively. This study analyzes electronic service quality, order fulfillment bottlenecks, and complaint management at Nasi Gerilya Restaurant on GrabFood in Medan City. We examine 4,584 GrabFood customer reviews, observe kitchen queue operations, and interview nine key informants across management, kitchen staff, cashiers, IT consultants, consumers, and competing restaurants. The restaurant maintains strong customer satisfaction through consistent food taste, generous meat portions, and responsive dish replacements. However, peak lunch hours create operational bottlenecks because dine-in customers, takeaway buyers, and delivery couriers crowd a single service counter. Packaging omissions, wrong food variants, and soup spillage cause recurring customer complaints. We map the order fulfillment process through a service blueprint and identify five operational failure points across e-service quality dimensions. We recommend a two-stage fail-safe packaging checklist, a separate courier dispatch station, and tamper-evident packaging to eliminate fulfillment errors and protect customer retention.

Keywords: e-service quality, complaint handling, GrabFood, service operations, service blueprint.

INTRODUCTION

Online food delivery has become a primary commercial channel for urban restaurants in post-pandemic Indonesia (Elverda & Sudarsono, 2025). Urban consumers choose food delivery apps for convenience and fast doorstep delivery (Chaffey & Ellis-Chadwick, 2022). For food and beverage small enterprises, platform partnerships expand customer reach without expanding physical dining space (Kotler et al., 2023). Food delivery platforms connect culinary merchants to large consumer bases through algorithmic matching and integrated payment gateways (Pigatto et al., 2017).

However, digital merchants face unique operational challenges. On delivery platforms, customers evaluate service quality through app responsiveness and order accuracy rather than dining room hospitality (Parasuraman et al., 2005; Zeithaml et al., 2002). Four core dimensions define this electronic service quality: fulfillment accuracy, system efficiency, menu availability, and dispute resolution (Parasuraman et al., 2005). Unlike dine-in restaurants where staff can resolve mistakes tableside, fulfillment errors on digital platforms immediately damage customer ratings (Santoso & Hidayat, 2023). These operational defects include omitted condiments, swapped food items, and extended driver waiting times. Public customer reviews display these failures directly to prospective buyers.

Nasi Gerilya is a Padang restaurant in Medan City operating since September 2024. The restaurant generates approximately 50% of its daily transactions from GrabFood. Nasi Gerilya maintains a strong online reputation with an average rating of 4.7 out of 5.0 across 4,584 reviews. The restaurant also achieved GrabFood *Rising Star* awards in 2025 and 2026. However, high transaction volumes cause severe congestion during lunch hours between 11:30 and 13:30 WIB. A single frontline counter processes dine-in orders, walk-in takeaway sales, and online driver pickups. Customer reviews reflect recurring complaints regarding missing condiments, swapped dishes, and driver waiting delays.

Existing literature predominantly examines consumer adoption drivers of food delivery applications (Putri & Rahayu, 2022). Few empirical studies investigate operational fulfillment bottlenecks inside restaurant kitchens. This study examines e-service quality dimensions at Nasi Gerilya on GrabFood. We identify specific failure points during order preparation and propose an operational service recovery framework.

LITERATURE REVIEW

Parasuraman et al. (2005) developed the e-service quality framework to evaluate online transactions across four core dimensions: efficiency, fulfillment, system availability, and privacy. Efficiency measures user interface speed and ease of navigation. Fulfillment covers delivery timeliness and order accuracy. System availability describes catalog accuracy and technical uptime. Privacy protects customer personal data. On food delivery platforms, fulfillment accuracy represents the most visible dimension to end consumers (Santoso & Hidayat, 2023).

In food services, order fulfillment connects the digital order to the physical food delivery (Wirtz & Lovelock, 2024). The service marketing 7P mix emphasizes processes, people, and physical evidence (Kotler et al., 2023). Physical evidence includes menu photos, item descriptions, receipt labels, and tamper-evident packaging. When delivered food does not match the app listing, customers experience a service failure. Quick staff responses, item replacements, and apologies restore customer trust (Smith et al., 1999; Wirtz & Lovelock, 2024). In contrast, slow or defensive merchant responses reduce customer return rates.

Physical service environments also shape customer and employee behavior. Bitner (1992) established the servicescape framework to explain how spatial layout, equipment placement, and ambient conditions affect operational flow. In multi-channel restaurants, physical counter crowding creates direct friction between walk-in patrons and delivery couriers. To prevent service errors during peak transaction velocity, service operations require mistake-proofing mechanisms. Chase and Stewart (1994) formulated fail-safe service design (poka-yoke) to eliminate human errors during order preparation. In food packaging, structured checklists ensure complete meal assembly before dispatch.

METHODS

This study applies a descriptive qualitative single-case design (Yin, 2018). We investigate order fulfillment operations at Nasi Gerilya Restaurant in Medan City. The single-case approach provides deep empirical access to kitchen processes, physical counter workflows, and digital review telemetry.

Informant Selection

We selected nine key informants through purposive sampling across management, kitchen operations, frontline cashiers, IT consultants, active platform consumers, and competing restaurants. Informants possess direct knowledge of GrabFood store operations, meal packaging, and multi-channel queuing. Table 1 displays the informant matrix, assigned informant codes (K01–K09), and operational competencies.

Table 1. Key Informant Matrix and Operational Competencies

No.	Code	Informant	Role / Position	Operational Competency and Scope
1	K01	Richard	Business Owner	Strategic channel mix, quality control, and POS software
2	K02	Ririn	Frontline Leader & Cashier	GrabFood POS, queue management, and order mediation
3	K03	Bella	Kitchen Assembler	Meal assembly, ingredient portioning, and box packing
4	K04	Devin	Business Consultant (APINDO)	Store layout advisory, queue flow, and process audits
5	K05	Alip	Active Consumer	Order accuracy, meal completeness, and missing gravy issues
6	K06	Reyza	Active Consumer	Promotion transparency, meat portions, and delivery speed
7	K07	Ojan	Active Consumer	Student meal bundles, value perception, and food taste
8	K08	Ken	Staff, RM Pondok Krakatau	Competitor packaging, driver queues, and stockout delays
9	K09	Asiong	Staff, RM Istana Krakatau	Competitor dispatch routines and dine-in vs delivery flows

Data Collection

We collected empirical data through four distinct methods:

1. **Digital Content Analysis**: We examined 4,584 GrabFood customer reviews retrieved from the merchant portal on June 13, 2026, capturing star ratings, customer text comments, and merchant operational replies.
2. **In-Depth Interviews**: We interviewed nine informants (K01–K09) using semi-structured protocols covering order preparation, packaging checks, driver handovers, and complaint resolution.
3. **Direct Observation**: We observed receipt printing, meal assembly, packaging routines, and courier handovers during peak lunch hours (11:30–13:30 WIB) across weekday and weekend service cycles.
4. **Document Audits**: We audited GrabFood menu architectures, physical store receipts, 19 promotional voucher terms, platform commission statements, and kitchen stockout registers.

Data Analysis

We analyzed qualitative data using the interactive model of Miles et al. (2014). This model includes data condensation, matrix displays, and conclusion drawing. We transcribed all audio recordings verbatim, extracted customer reviews into structured digital matrices, and systematically compared interview transcripts with platform telemetry and observational field notes. Thematic coding mapped operational friction points to the four e-service quality dimensions of Parasuraman et al. (2005).

Data Trustworthiness

To ensure methodological rigour and analytical validity, this study implemented the four-dimension trustworthiness criteria formulated by Lincoln and Guba (1985): credibility, transferability, dependability, and confirmability. Table 2 details the operationalization of each trustworthiness dimension, mapping the empirical data sources, informant roles, and audit trail verification procedures.

Table 2. Qualitative Trustworthiness Framework and Data Audit Trail

Dimension	Operationalization in Case	Data Sources & Informants	Verification & Audit Trail Procedure
Credibility	Prolonged field engagement and multi-source triangulation	4,584 GrabFood reviews; informants K01–K04 (internal) and K05–K07 (consumers)	Cross-referencing digital complaint logs against kitchen assembly observations; member-checking interview summaries with owner (K01) and cashier (K02).
Transferability	Dense contextual description of multi-channel restaurant operations	Storefront parameters; field layout notes; competitor staff (K08, K09)	Detailed documentation of kitchen workflow, counter geometry, 50% platform revenue reliance, and Padang cuisine packaging characteristics.
Dependability	Systematic operational audit trail and procedural stability	Audio recordings, verbatim transcripts, digital review sheets, POS logs	Maintaining unified inquiry audit trail; standardized coding schemes mapped to E-S-QUAL dimensions; peer debriefing with academic advisors.
Confirmability	Reflexive bracketing of researcher bias and neutral evidence tracing	Multi-informant triangulation (K01–K09); raw customer quotes; POS logs	Grounding all inferences in verbatim interview excerpts and unedited customer reviews; cross-checking internal staff claims against external competitor practices.

The systematic combination of digital review logs, direct time-and-motion observation, and multi-tier informant interviews provides robust triangulation across data sources and methodological techniques. This protocol prevents anecdotal bias and ensures that operational diagnoses reflect verified fulfillment dynamics.

RESULTS

Fieldwork reveals that Nasi Gerilya maintains high customer satisfaction on GrabFood. However, operational bottlenecks slow down order preparation and dispatch.

Storefront Audit

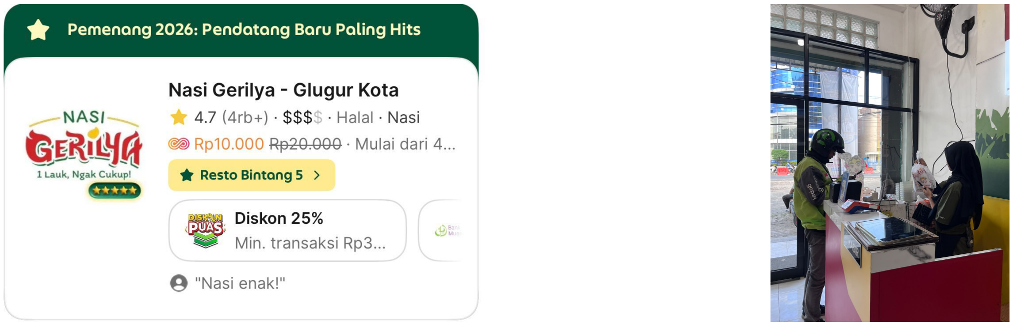
Table 3 outlines the audited storefront parameters. The storefront displays an aggregate rating of 4.7 out of 5.0 from 4,584 reviews and GrabFood *Rising Star* badges. However, several popular items remain out of stock on the app.

Table 3. Operational Storefront Parameters Audited on GrabFood

No.	Storefront Parameter	Empirical Observation Findings (June 13, 2026)
1	Outlet Identity & Rating	Nasi Gerilya - Glugur Kota; Rating 4.7 / 5.0 (4,584 verified customer reviews)
2	Recognition Badges	GrabFood *Rising Star* Winner 2025 & 2026 (5-Star Restaurant Designation)
3	Promotional Campaigns	19 active promotional vouchers (bank, discounts, delivery subsidies)
4	Signature Dishes	Beef Rendang, Spiced Fried Chicken Rice, and Creamy Pop Chicken Rice
5	Price Range	Rp1,100 (additional curry sauce) to Rp72,000 (premium multi-dish packages)
6	Catalog Stockouts	6 items marked *Sold Out* (premium fish head and packaged sambals)

When customers order dishes that the kitchen has run out of, staff must call buyers to change menus or cancel orders.

Figure 1 displays the digital storefront alongside the physical cashier counter. Figure 1b shows a single counter handling dine-in guests, takeaway buyers, and delivery drivers. This layout creates severe congestion



(a). Digital Storefront and Rating Badges

(b). Frontline Cashier Workstation

Figure 1. Digital Interface and Physical Fulfillment Workstation at Nasi Gerilya

during peak hours.

To examine catalog availability friction, Table 4 lists the six menu items marked unavailable during platform observation.

Table 4. Audit of Unavailable Menu Items on GrabFood Storefront

No.	Menu Item Name	Price (Rp)	Operational Stockout Category
1	Gulai Merah Kepala Ikan (Size L)	275,000	Premium fish item; limited daily procurement
2	Gulai Merah Kepala Ikan (Size M)	248,050	Premium fish item; limited daily procurement
3	Nasi Padang Gembung Kuring Bakar	47,850	Fresh fish dish; exhausted before lunch peak
4	Nasi Padang Ayam Pop Creamy	46,000	Core signature dish; high customer demand
5	Sambal Cumi Nagih Kemasan	38,500	Packaged retail item; batch production run delay
6	Sambal Tuna Asap Keumamah Aceh	38,500	Packaged retail item; batch production run delay

Table 4 demonstrates that stockouts affect both high-ticket specialty seafood and best-selling core dishes. Unsynchronized inventory causes immediate customer disappointment when users attempt to reorder signature items.

Promotional Schemes

Table 5 details the 19 active promotional vouchers on GrabFood. Commercial bank partnerships dominate promotions and attract new buyers. However, the restaurant does not sync kitchen inventory with the POS terminal. As a result, the digital menu displays exhausted dishes.

Table 5. Distribution of Active Promotional Campaigns and Inventory Stockouts

No.	Promotional Mechanism	Items	Operational Terms and Merchant Objective
1	Commercial Bank Co-Marketing	11	Minimum spend Rp40k–Rp100k; discounts Rp4k–Rp25k
2	Merchant-Funded Percentage Off	2	17%–25% discount (capped at Rp50k–Rp100k)
3	Delivery Subsidization	3	Rp3,000–Rp8,000 discount; offsets customer delivery cost
4	Dish-Level Deduction	1	Direct Rp4,500 price reduction applied to Minced Meat Curry
5	Collaborative Group Order	1	5% discount for 3–6 joint orders
6	Stockout Friction Points	6	Snapper Head Curry and sambal jars *Sold Out*

Bank promotions attract budget-conscious buyers and drive lunch order surges. These surges quickly deplete popular dishes in the kitchen. The cashiers must then call customers manually to negotiate meal replacements.

Menu and Pricing Structure

Table 6 presents menu categories and price distributions. Menu tiers range from vegetable rice at Rp27,500 to mutton curry at Rp72,000. À la carte proteins cost between Rp27,000 and Rp67,500.

Table 6. Menu Architecture and Pricing Tiers on GrabFood Storefront

No.	Menu Category	Count	Price Range (Rp)	Representative Core Items
1	Complete Rice Packages	26	27,500 – 72,000	Rendang Sapi, Ayam Pop, Kari Kambing
2	Additional Side Dishes	7	7,700 – 27,500	Jengkol Rendang, Telur Dadar, Keripik Kentang
3	À La Carte Proteins	23	27,000 – 67,500	Gulai Cincang, Ayam Bakar, Udang Goreng
4	Fresh Table Sambals	2	17,500	Sambal Hijau Lado Mudo, Sambal Merah
5	Bottled Specialty Sambals	6	38,500	Sambal Andaliman, Cumi Nagih, Teri Medan
6	Gravy Supplements	2	1,100 – 2,500	Kuah Gulai Kuning, Kuah Campur

Table 6 shows that complete rice packages dominate the catalog with 26 SKUs. Each box includes rice, meat, vegetables, green chili, and curry gravy. Packing many separate items increases the risk of omitting condiments during lunch rush.

Platform Price Mark-Up

To evaluate the economic factors driving consumer grievance intensity, Table 7 compares in-store dine-in prices with GrabFood digital prices across core signature dishes.

Table 7. Comparative Pricing Audit: Dine-In vs. GrabFood Platform Mark-Up

No.	Core Menu Item	Dine-In (Rp)	GrabFood (Rp)	Mark-Up	Operational Cost Driver and Packaging Allocation
1	Nasi Sayur Polos	22,000	27,500	+25.0%	Platform fee (20%) and partitioned meal box
2	Nasi Ayam Goreng Bumbu	36,000	45,000	+25.0%	Platform fee (20%) and grease-proof box packaging
3	Nasi Ayam Pop Creamy	37,000	46,000	+24.3%	Platform fee (20%) and separate sauce cup
4	Nasi Rendang Sapi	45,000	56,000	+24.4%	Platform fee (20%) and separate gravy cup
5	Nasi Kikil Tunjang Gulai	55,000	68,000	+23.6%	Platform fee (20%) and double-lined takeout tub
6	Nasi Kari Kambing	58,000	72,000	+24.1%	Platform fee (20%) and reinforced soup bowl

Table 7 demonstrates a consistent 23%–25% pricing mark-up on GrabFood across all core packages. This mark-up covers the standard 20% merchant platform commission and dedicated takeout packaging materials. Field interviews reveal that paying a substantial price premium heightens consumer quality expectations. When online consumers pay Rp56,000 for beef rendang that costs Rp45,000 in-store, an omitted condiment pouch is perceived not as a minor oversight, but as an unacceptable fulfillment failure justifying an immediate 1-star penalty.

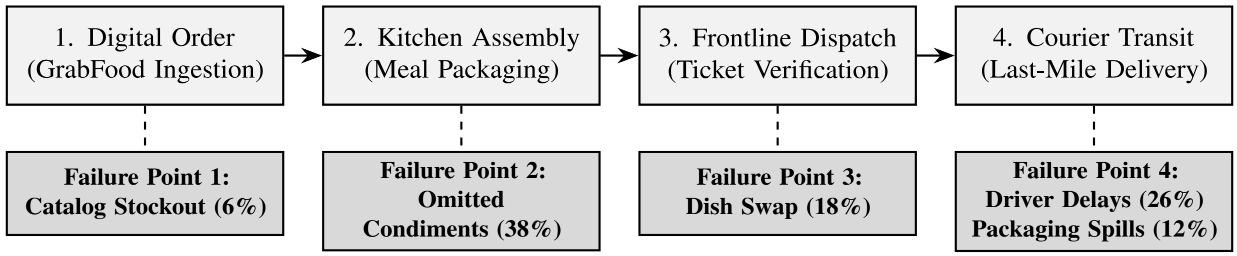


Figure 2. Service Blueprint of Order Fulfillment Journey and Operational Failure Points

Fulfillment Service Blueprint

To analyze how order fulfillment breaks down during peak velocity, we mapped the operational workflow into a service blueprint. Figure 2 illustrates the four physical fulfillment stages and their corresponding failure points.

Figure 2 shows that operational errors concentrate at specific handover boundaries. The kitchen assembly stage generates condiment omissions, while the cashier dispatch stage causes variant mix-ups due to visually identical paper receipts.

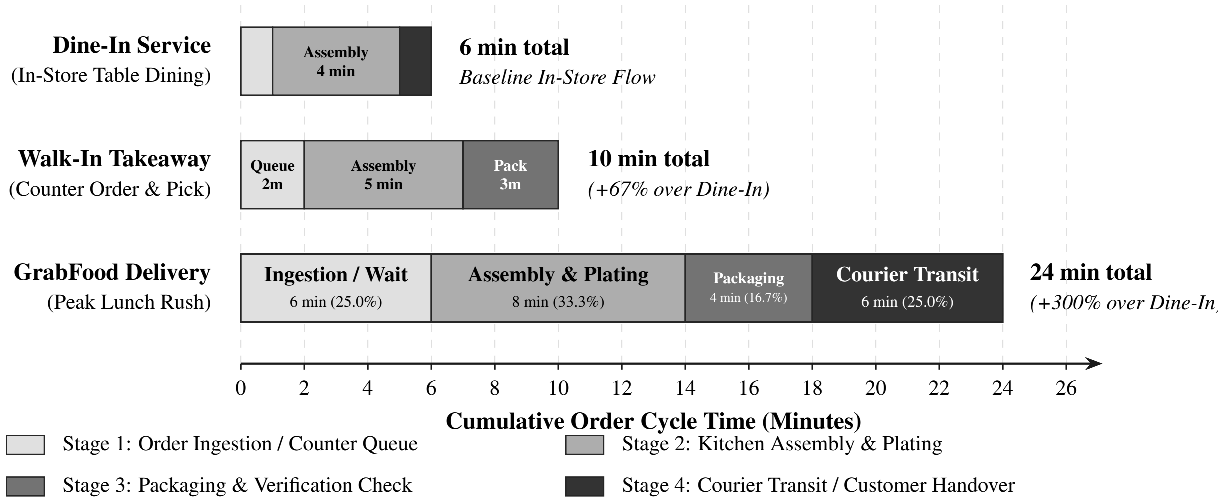
Queue Dynamics and Cycle Times

Multi-channel food service operations experience queue friction when online and offline customer flows intersect at an unsegregated service counter. To quantify how order velocity degrades under multi-channel contention, we conducted time-and-motion observations tracking cumulative order cycle times (W) across distinct fulfillment channels during the peak lunch rush (11:30–13:30 WIB). In accordance with queueing theory and Little's Law ($L = \lambda W$), sudden surges in digital order arrival rates (λ) disproportionately inflate order cycle times when service capacity remains fixed.

Figure 3 illustrates the cumulative order cycle time profiles decomposed across four sequential fulfillment stages: Stage 1 (Order Ingestion and Counter Queue), Stage 2 (Kitchen Assembly and Plating), Stage 3 (Packaging and Verification Check), and Stage 4 (Courier Transit or Tableside Handover).

As depicted in Figure 3, dine-in table orders achieve the fastest fulfillment velocity, requiring an average of only 6 minutes from order placement to tableside presentation (1 minute for POS logging, 4 minutes for kitchen plating, and 1 minute for table serving). Walk-in takeaway customers experience an average cycle time of 10 minutes (+67% higher than dine-in), driven by manual counter queuing (2 minutes) and takeout box packing (3 minutes). In sharp contrast, GrabFood delivery orders during the peak lunch window reach an average cycle time of 24 minutes, representing a 300% increase over the baseline dine-in service.

To evaluate the operational bottlenecks across service stages, Table 8 provides an empirical breakdown of mean durations, percentage allocations, and observed friction mechanisms across service channels.



Operational Cycle Time Dynamics (GrabFood Peak 24 min): Internal Kitchen Operations = 12 min (50.0%: Assembly 8m + Packaging 4m) | Platform & Logistics Latency = 12 min (50.0%: Ingestion/Driver Wait 6m + Courier Transit 6m). Peak lunch rush doubles kitchen assembly duration from 4 to 8 minutes due to cross-channel queue contention at the single frontline counter.

Figure 3. Comparative Order Cycle Time Breakdown Across Service Channels During Peak Lunch Rush

Table 8. Temporal Distribution of Order Cycle Times and Channel Bottlenecks (11:30–13:30 WIB)

Channel / Order Mode	Ingest	Assembly	Pack	Transit	Total	Primary Operational Bottleneck & Queue Friction
Dine-In (In-Store Table)	1.0 m	4.0 m	–	1.0 m	6.0 m	Direct verbal ticket transmission; minimal queue friction
Walk-In Takeaway (Counter)	2.0 m	5.0 m	3.0 m	–	10.0 m	Counter payment queue; single-staff manual box assembly
GrabFood (Off-Peak Window)	2.0 m	5.0 m	2.5 m	4.5 m	14.0 m	Immediate driver matching; smooth kitchen throughput
GrabFood (Peak Lunch Rush)	6.0 m	8.0 m	4.0 m	6.0 m	24.0 m	Driver clustering at single POS; batching delay; bag sealing

Table 8 reveals that for GrabFood peak-hour orders, internal kitchen operations account for 12.0 minutes (50.0% of total cycle time, split into 8.0 minutes for assembly and 4.0 minutes for packaging), whereas external platform and logistics latency contributes another 12.0 minutes (50.0%, comprising 6.0 minutes of digital ingestion and driver arrival wait, plus 6.0 minutes of last-mile road transit). The peak lunch rush doubles kitchen assembly time from 4.0 to 8.0 minutes because kitchen staff must batch digital orders while concurrently serving visible dine-in patrons. In addition, the 4.0 minutes required for packaging separate condiment pouches under peak-hour physical constraints creates the operational strain that leads to omission errors.

Customer Grievance Analysis

We coded 4,584 reviews and identified five recurring operational failures. Table 9 explains the root causes observed during restaurant service encounters.

Table 9. Classification of Operational Service Failures Across E-S-QUAL Dimensions

No.	Failure Category	E-S-QUAL Dimension	Share	Root Cause and Mechanism Observed in Fieldwork
1	Missing Condiments	Fulfillment	38%	Curry gravy, sambal pouches, or cutlery omitted during lunch rush
2	Driver Waiting Delays	Efficiency	26%	Couriers wait 15–25 minutes as staff prioritize in-store patrons
3	Incorrect Menu Variant	Fulfillment	18%	Swapped meal boxes caused by visually similar 4-digit order tickets
4	Packaging Spillage	Physical Evidence	12%	Loosely fastened gravy pouches tear inside carrier bags in transit
5	Real-Time Stockout	Availability	6%	Depleted items remain active digitally, forcing phone cancellations

Fieldwork observations in Table 9 confirm that errors do not come from cooking mistakes. Instead, counter crowding and the absence of a packing checklist cause these failures.

As shown in Figure 4, missing items (38%) and driver waiting times (26%) account for 64% of all customer complaints. These two operational issues stem directly from queue bottlenecks and single-checker packing.

Grievance Taxonomy and Rating Impact

To examine how fulfillment failures affect customer sentiment, we audited the repository of 4,584 GrabFood customer reviews. On food delivery platforms, consumer reviews represent unstructured digital feedback where text sentiment aligns with numerical ratings. Across the 4,584 reviews, the store maintains an aggregate rating of 4.70 out of 5.0, reflecting a dominant base of satisfied customers: 3,854 five-star reviews (84.07%) and 422 four-star reviews (9.21%), totaling 93.28% positive evaluations. Critical reviews ($\leq$ 3 stars) account for only 6.72% of total volume (308 reviews, comprising 102 three-star, 64 two-star, and 142 one-star ratings). While positive reviews praise culinary attributes using descriptors such as *"gurih"* (savory), *"porsi besar"* (generous portion), and *"empuk"* (tender), the 308 critical reviews concentrate heavily around operational fulfillment failures.

Across these 308 critical reviews, we systematically extracted and coded 260 explicit operational grievance mentions into semantic clusters (the remaining 48 reviews contained non-operational feedback or star-only ratings without text). For each keyword cluster, Table 10 details the frequency of occurrence, corresponding share among recorded grievances, rating distribution (1*, 2*, 3*), mean rating, rating penalty relative to the baseline 4.70 store rating, and the underlying operational failure mechanism.

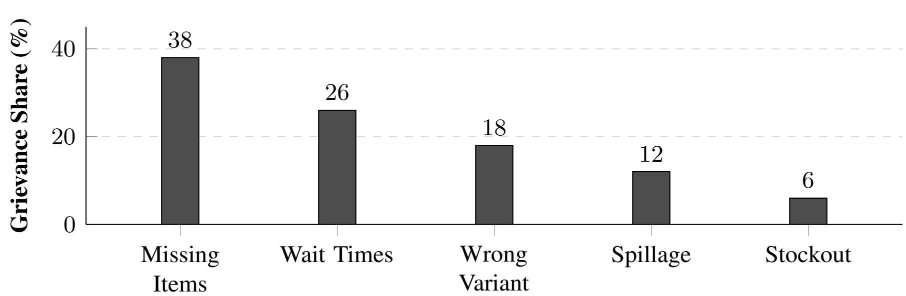


Figure 4. Quantitative Distribution of Customer Grievance Categories

Table 10. Linguistic Keyword Taxonomy and Valence Impact of Customer Grievances

Keyword Cluster (ID / EN)	Count	Share	Stars ($1^*/2^*/3^*$)	Rating (Penalty)	Operational Failure Mechanism
"kuah", "bumbu" (curry gravy)	58	22.3%	34 / 16 / 8	1.55 (-3.15)	Curry gravy omitted; dry meal experience
"sambal", "lado" (chili cup)	41	15.8%	22 / 12 / 7	1.63 (-3.07)	Sambal pouch forgotten; loss of flavor
"salah menu" (swapped dish)	47	18.1%	39 / 6 / 2	1.21 (-3.49)	Ticket code swapped; wrong dish received
"lama", "antri" (late delivery)	44	16.9%	11 / 12 / 21	2.23 (-2.47)	Courier wait $>$ 20 min; late delivery
"tumpah", "bocor" (gravy spill)	31	11.9%	15 / 9 / 7	1.74 (-2.96)	Single-knot sauce pouch torn in transit
"dingin", "anyep" (loss of heat)	24	9.2%	3 / 6 / 15	2.50 (-2.20)	Loss of food heat from delivery transit
"habis", "batal" (menu stock-out)	15	5.8%	11 / 3 / 1	1.33 (-3.37)	Depleted item ordered; cashier cancellation
Total Grievance Mentions	**260**	**100.0%**	**135 / 64 / 61**	**1.69** (-3.01)	**Coded across 308 critical reviews**

The empirical findings in Table 10 show that omitted condiments (*kuah* and *sambal*, totaling 99 mentions or 38.1%) and swapped variants (47 mentions or 18.1%) account for over half of all customer grievances, matching the aggregate failure distribution identified in Figure 4. Furthermore, wrong items and omitted condiments trigger the sharpest valence penalties. Receiving a swapped menu variant incurs a mean rating drop of -3.49 points (yielding an average rating of only 1.21 out of 5.0, where 39 out of 47 reviewers assigned 1 star), while omitted curry gravy drops the rating by -3.15 points (to 1.55 out of 5.0, with 34 out of 58 reviewers assigning 1 star). In customer feedback, these low ratings are frequently paired with explicit comments noting that while food taste is satisfactory, omitting essential condiments spoils the meal experience. This confirms that peripheral components carry disproportionate weight in consumer evaluations.

Staff and Management Perspectives

To understand the underlying causes of packaging omissions and queue congestion, we conducted in-depth interviews with internal restaurant personnel. Table 11 compiles representative interview statements from management, kitchen staff, cashiers, and IT consultants.

Table 11. Thematic Coding of Operational Fieldwork Interviews

No.	Informant	Operational Area	Verbatim Fieldwork Interview Insight
1	Richard	Quality Control	*"Consistency in taste is non-negotiable. If spices fail our standard, we discard the batch. But delivery packing requires stricter oversight."*
2	Bella	Kitchen Assembly	*"During peak rush, packing gets frantic. Items packed in separate plastic pouches, like curry gravy and sambal, are vulnerable to omission."*
3	Ririn	Cashier & Dispatch	*"Four-digit order codes on printed tickets often look identical. If a driver mentions the wrong code, meal boxes can get swapped."*
4	Devin	Queue Architecture	*"Dine-in, takeaway, and delivery couriers crowd one counter. Without physical lane separation, driver waiting times increase rapidly."*

Table 11 confirms that internal staff recognize the operational friction points. The kitchen leader identifies separate gravy pouches as the primary omission risk, while the cashier attributes dish swaps to visually indistinguishable POS receipts.

Customer Review Evidence

Customer reviews show a sharp contrast between delicious food and fulfillment errors. Table 12 presents verbatim customer reviews from the GrabFood storefront.

Table 12. Representative Verbatim Customer Reviews from GrabFood Storefront

No.	Reviewer ID	Rating	Valence	Verbatim Review Extract and Operational Context
1	Raymond	5/5	Positive	*"The pop chicken is creamy, tender, and seasoned to perfection."*
2	Wie W.	5/5	Positive	*"Weekly family order; generous portions, clean boxes, requests met."*
3	Talia	1/5	Negative	*"Ordered chicken rendang, but received fried chicken instead."*
4	Hendra	3/5	Neutral	*"Ordered fish but received tilapia; merchant issued formal apology."*
5	Siyaga	3/5	Neutral	*"Food is delicious, but staff omitted extra curry; responsive merchant."*
6	Albert P.	1/5	Negative	*"Staff omitted crispy skin crackers; merchant apologized for oversight."*

Table 12 shows that consistent taste earns 5-star ratings. However, missing gravy or wrong dishes immediately trigger 1-star ratings. These operational errors undermine customer retention.

Unit Economics of Service Failure

To evaluate the financial implications of operational service failures, we modeled the unit economics of order fulfillment and service recovery. Culinary merchants on on-demand delivery platforms operate within constrained profit margins shaped by standard platform commissions (typically 20% of gross merchandise value) and dedicated takeout packaging costs. When an order encounters a fulfillment defect, the merchant must either absorb substantial operational recovery expenses or suffer platform dispute clawbacks.

We analyzed the unit economics for a representative high-volume signature dish, Nasi Rendang Sapi, priced at Rp56,000 on GrabFood (reflecting a 24.4% platform mark-up over the in-store dine-in price of Rp45,000). Under a flawless fulfillment scenario, the gross order value yields a merchant payout of Rp44,800 after the 20% platform fee (Rp11,200). Deducting food ingredient costs (Cost of Goods Sold, estimated at 45% of dine-in value, or Rp20,250) and takeout packaging materials (Rp2,500) leaves a net operating contribution of Rp22,050 per box, representing a healthy 39.38% net merchant margin.

However, when a service failure occurs, this margin erodes significantly depending on the recovery mechanism employed. Table 13 details the financial trajectory across five fulfillment scenarios: baseline flawless execution, minor omission resolved via on-demand courier resend, minor omission compensated via digital voucher, critical failure leading to platform dispute refund, and critical failure rectified via emergency dish replacement with express courier dispatch.

Table 13. Unit Economics of Order Fulfillment and Service Recovery Margin Erosion (Nasi Rendang Sapi)

Fulfillment Scenario	Price	Payout	Base Cost	Recovery	Net Profit	Margin	Erosion
1. Flawless Order (Baseline)	56,000	44,800	22,750	0	+22,050	39.4%	Baseline
2. Omission + Express Courier Resend	56,000	44,800	22,750	19,500	+2,550	4.6%	-88.4%
3. Omission + Digital Store Voucher	56,000	44,800	22,750	10,000	+12,050	21.5%	-45.4%
4. Swapped Dish + Full Re-Dispatch	56,000	44,800	22,750	40,750	$-18,700$	-33.4%	-184.8%
5. Unresolved Defect (Portal Refund)	56,000	0	22,750	0	$-22,750$	-40.6%	-203.2%

As demonstrated in Table 13, dispatching an on-demand courier to deliver a forgotten gravy pouch (Scenario 2) incurs Rp16,500 in third-party courier delivery fees plus Rp3,000 in additional packaging and sauce preparation costs, totaling Rp19,500 in recovery expenditure. This operational response reduces net profit from Rp22,050 to just Rp2,550, producing an 88.4% margin erosion on the transaction.

When disputes remain unresolved, such as when a swapped dish or spillage occurs (Scenario 5), the delivery platform automatically issues a full refund to the customer and rescinds the merchant's Rp44,800 payout. In this situation, the merchant incurs an outright operating loss of Rp22,750 on the meal, losing 100% of sunk food and packaging expenditures. Alternatively, when management attempts immediate physical restitution by cooking a fresh meal and hiring an instant courier (Scenario 4), the cumulative fulfillment cost surges to Rp63,500 against a net payout of Rp44,800, generating a net operating deficit of Rp18,700.

These quantitative dynamics demonstrate that service recovery in digital delivery ecosystems cannot be treated as an acceptable recurring operating overhead. Given that 38% of customer grievances involve omitted condiments and 18% involve swapped variants, an operational defect rate of even 5% across 150 daily orders would erode up to Rp150,000 to Rp300,000 in daily operating profits. This economic reality establishes that mistake-proofing mechanisms (poka-yoke) and frontline verification protocols are essential investments for commercial viability.

Cross-Merchant Comparison

To assess whether fulfillment errors represent an idiosyncratic problem of Nasi Gerilya or an industry-wide challenge, we interviewed operational staff from two competing Padang restaurants on GrabFood. Table 14 compares operational practices across merchants.

Table 14. Comparative Triangulation Across Competing Padang Restaurants on GrabFood

No.	Establishment	Online Share	Primary Service Bottlenecks	Fulfillment Recovery Practice
1	Nasi Gerilya	∼50%	Missing gravy (38%) and driver wait (26%)	Instant replacement with merchant apology
2	RM Pondok Krakatau	∼50%	Driver delays and unsynced stockouts	Missing item redelivery via courier
3	RM Istana Krakatau	∼20%	Omitted side dishes and lunch rush	Direct manual checker inspection

Table 14 demonstrates that multi-channel queue crowding and condiment omissions affect competing food merchants across Medan. Therefore, packaging errors and courier wait times represent systemic operational challenges in online food delivery.

DISCUSSION

These empirical findings substantiate the core propositions of Parasuraman et al. (2005) and Wirtz and Lovelock (2024). On on-demand delivery platforms, fulfillment accuracy represents a foundational hygiene factor rather than a secondary differentiator. Customers consistently praise culinary attributes, such as portion volume and authentic Padang seasoning, with 5-star ratings. However, fulfillment failures directly penalize these ratings. When staff omit curry sauce or send the wrong dish, customers immediately assign 1-star reviews. This asymmetric penalty demonstrates that culinary satisfaction cannot compensate for fulfillment errors in digital environments (Santoso & Hidayat, 2023).

Theoretical Analysis

To conceptualize the cognitive mechanisms driving extreme rating drops, our findings integrate Expectation-Disconfirmation Theory (EDT) (Oliver, 1980) with the theory of Asymmetric Attribute Performance (Anderson & Sullivan, 1993; Mittal et al., 1998). EDT posits that customer satisfaction is determined by the psychological gap between prior expectations and perceived performance. Positive disconfirmation occurs when performance exceeds expectations, whereas negative disconfirmation occurs when performance falls short. In food delivery services, however, this psychological mechanism operates with pronounced structural asymmetry.

Johnston (1995) classified service quality attributes into satisfiers and dissatisfiers (hygiene factors). Satisfiers are attributes that enhance satisfaction when fulfilled but do not cause intense dissatisfaction when absent. Conversely, dissatisfiers or hygiene factors are baseline operational attributes that customers take for granted; their presence merely prevents dissatisfaction, but their absence triggers negative disconfirmation. In Indonesian culinary culture, particularly for Padang cuisine, rice meals are fundamentally consumed with copious curry gravy (*kuah gulai*) and sambal. While generous protein portions and authentic spice blends serve as core satisfiers, the inclusion of complete sauces and condiments functions as an absolute hygiene factor.

When a restaurant successfully delivers a well-spiced rendang with complete gravy, customers experience neutral or mild positive confirmation, awarding 4 or 5 stars. However, when the kitchen omits the gravy pouch, negative disconfirmation occurs because the meal becomes dry without sauce. Under the asymmetric impact model of Mittal et al. (1998), negative performance on a hygiene attribute exerts a greater impact on overall satisfaction than an equivalent positive performance on a satisfier attribute. As documented in our empirical valence audit (Table 10), the absence of a Rp1,100 condiment pouch diminishes the customer's perceived value of an entire Rp56,000 meal package, resulting in a 1-star review.

This cognitive asymmetry is reinforced by the platform pricing mark-up. Because consumers pay a 24.4% price premium over physical dine-in prices (Table 7), their initial performance expectations are higher than in-store diners. At this higher expectation threshold, customers show lower tolerance for execution defects. Unlike physical dining rooms where customers can immediately signal a missing sauce to waitstaff, digital delivery encounters are separated by physical distance and third-party couriers. Once the driver departs, the customer faces friction in obtaining redress, transforming operational oversights into customer dissatisfaction.

Digital Service Recovery Framework

When service failures occur, restaurants require structured recovery protocols to mitigate customer dissatisfaction, prevent negative reviews, and protect long-term retention (Smith et al., 1999; Tax et al., 1998). On digital platforms, service recovery is complicated by the presence of a third-party intermediary, where merchant response speed and perceived justice dictate whether a customer churns permanently (Davidow, 2003; Maxham & Netemeyer, 2002). Based on our empirical defect classification, we propose a Hierarchical Digital Service Recovery Framework.

Figure 5 illustrates the multi-tier operational decision tree, routing detected defects into three distinct severity levels with explicit escalation paths, procedural actions, target Service Level Agreements (SLAs), and closed-loop feedback governance.

As shown in Figure 5, the framework categorizes service failures into three hierarchical tiers: Level 1 (Minor Service Defect), Level 2 (Moderate Service Failure), and Level 3 (Critical Service Failure). Each tier corresponds to a specific operational profile and requires an aligned recovery response.

To operationalize this decision framework, Table 15 outlines the multi-tiered service recovery action matrix,

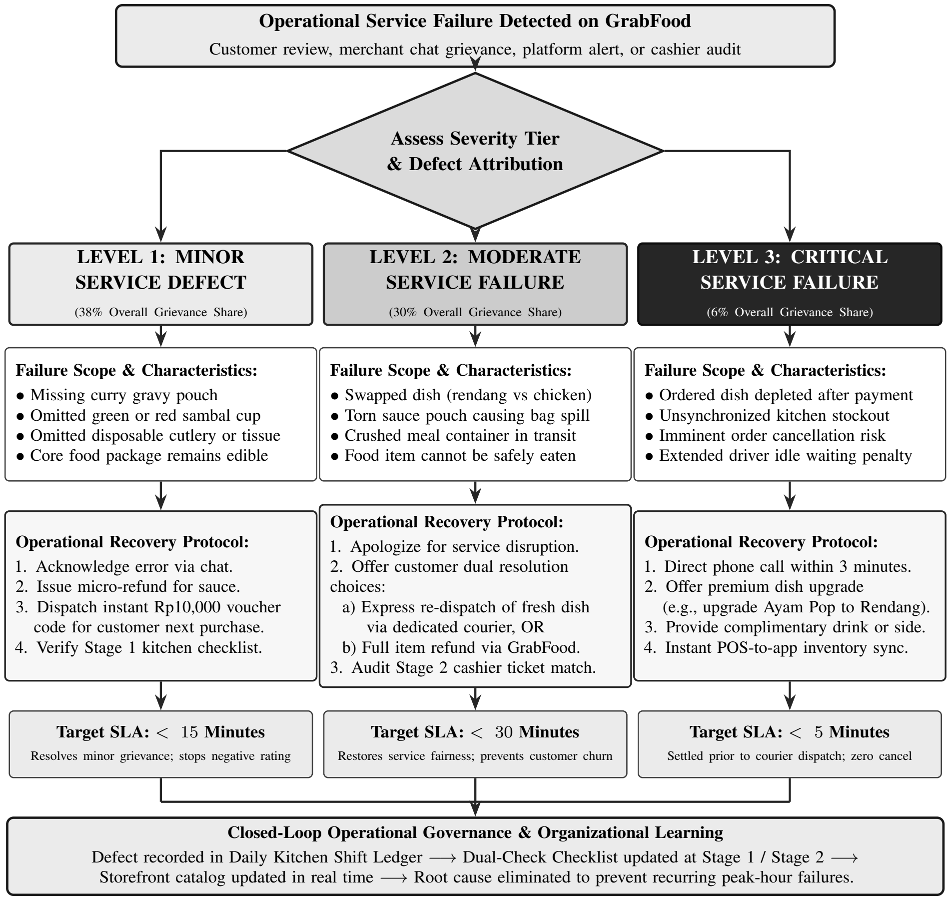


Figure 5. Hierarchical Service Recovery Decision Tree and Closed-Loop Escalation Protocol

detailing defect scopes, operational recovery actions, target resolution SLAs, and organizational feedback mechanisms across all three severity levels.

Table 15. Multi-Tiered Service Recovery Action Matrix and Operational SLAs

Severity Level	Defect Scope & Trigger	Operational Service Recovery Protocol	Target SLA	Governance & Feedback Loop
Level 1: Minor (38% share)	Missing gravy, forgotten sambal, omitted cutlery or napkins; core food intact	Send proactive in-app apology; process instant micro-refund for sauce value; credit customer with Rp10,000 digital voucher for next order	< 15 min	Log defect in Daily Kitchen Shift Ledger; inspect packer adherence to Stage 1 checklist
Level 2: Moderate (30% share)	Swapped food variants, torn sauce pouch, spillage inside bag; meal cannot be eaten	Offer customer choice: (a) express re-dispatch of fresh meal via on-demand courier, or (b) full platform refund; personal phone apology	< 30 min	Cashier audit of Stage 2 ticket matching; retrain staff on lid taping and tamper-evident sealing
Level 3: Critical (6% share)	Catalog stockout after payment; wrong address dispatch; driver wait > 25 min	Direct phone call within 3 minutes; offer premium menu upgrade (e.g. Ayam Pop to Rendang) with complimentary drink; instant catalog sync	< 5 min	Immediate POS inventory adjustment; disable depleted SKU on GrabFood portal; shift leader review

The operational matrix in Table 15 establishes clear boundaries between minor cosmetic defects and critical failures. By offering digital vouchers for Level 1 omissions, the merchant avoids the prohibitive Rp19,500 courier resend cost while preserving customer loyalty (Table 13). For Level 2 failures, providing an immediate choice between express redelivery and a full refund restores distributive and interactional justice (Smith et al., 1999). For Level 3 stockouts, resolving discrepancies before courier departure prevents order cancellations.

Queue dynamics also reveal operational decoupling challenges between physical and virtual service encounters (Bitner, 1992). The shared frontline counter creates direct interference between dine-in guests and delivery couriers. Cooks prioritize visible seated customers over digital tickets. This queue bias increases driver waiting times and causes courier friction.

To address these vulnerabilities, we structure three operational improvements:

Two-Stage Packaging Protocol

Packaging errors stem from relying on a single cook during lunch rush hours. The restaurant should implement a two-stage fail-safe checklist (Chase & Stewart, 1994), structured as an operational poka-yoke inspection routine. Table 16 outlines the procedural stages, control parameters, and physical mechanisms.

Table 16. Proposed Two-Stage Fail-Safe Quality Verification Checklist

Stage	Inspector	Control Parameter	Inspection Criterion	Poka-Yoke Mechanism
1	Kitchen Packer	Meal Completeness	Rice portion, main protein, and side vegetables	Physical ticket item check
1	Kitchen Packer	Condiments & Sauces	Curry gravy pouch, sambal pouch, and chili cup	Count of liquid pouches
1	Kitchen Packer	Eating Utensils	Spoon, fork, and hygiene tissue paper	Dedicated cutlery slot fill
2	Frontline Cashier	Ticket Matching	4-digit GrabFood code vs. driver app display	Verbal and visual code match
2	Frontline Cashier	Transit Security	Heat-resistant lid, taped gravy, and tamper seal	Tamper-evident seal adhesion

Under this protocol, Stage 1 guarantees meal integrity inside the kitchen, while Stage 2 prevents dispatching mismatched order bags to drivers.

Servicescape Decoupling

Driver waiting times (26%) occur because delivery couriers and dine-in patrons compete for frontline space. Physical counter crowding degrades service encounters for both groups (Bitner, 1992). To resolve this spatial friction, Figure 6 contrasts the existing counter layout with the proposed decoupled servicescape model.

As illustrated in Figure 6b, establishing a dedicated digital dispatch station decouples the physical service encounter. Couriers collect pre-bagged orders directly from a side hatch without entering the customer dining queue, reducing driver waiting times and dining room congestion.

Packaging and Inventory Controls

Packaging spillage causes 12% of recorded complaints. Branded tamper-evident stickers will secure food containers and reassure customers regarding transit hygiene (Kotler et al., 2023). Management must also synchronize POS inventory with the GrabFood merchant app. Real-time menu updates automatically disable depleted dishes, preventing out-of-stock cancellations.

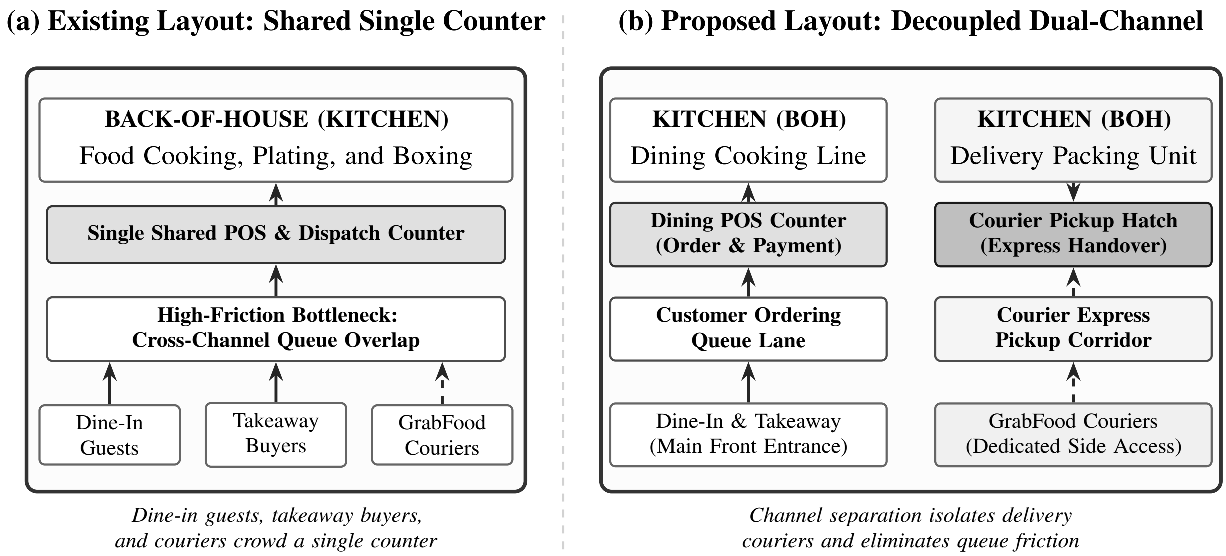


Figure 6. Servicescape Layout Schematic: Existing Bottleneck vs. Proposed Decoupled Counter

Managerial Implications

The empirical findings provide actionable, operationally grounded managerial guidance for small and medium food enterprises operating in multi-channel urban delivery ecosystems. Rather than relying on superficial marketing or continuous promotional discounting, restaurant managers must address root operational causes across four structured, capital-efficient operational pillars:

1. **Managing Asymmetric Quality Perception and Condiment Hygiene Factors**: Restaurant operators must recognize the cognitive asymmetry governing digital consumer evaluations (Oliver, 1980; Johnston, 1995). While kitchen staff often view side sauces, green chili cups, and curry gravy as low-cost incidental add-ons, digital delivery consumers treat these condiments as essential hygiene attributes. An omitted condiment pouch induces negative disconfirmation, diminishing the perceived utility of an otherwise well-prepared meal and triggering severe 1-star reviews (Table 10). Management must instill an organizational culture where condiment completeness is treated with the same operational discipline as main protein cooking. Condiments should be pre-portioned, sealed, and inventoried during pre-rush staging rather than ladled on-the-fly during peak lunch velocity.
2. **Physical Servicescape Decoupling with Low Capital Expenditure**: Counter crowding between dine-in patrons, walk-in takeaway buyers, and delivery couriers represents a primary driver of driver waiting delays and kitchen confusion (Bitner, 1992). Multi-channel restaurants can resolve this physical friction without prohibitive architectural renovation capital. Operators should reallocate existing counter geometry into two distinct operational streams (Figure 6b): an internal ordering lane for dining patrons and a dedicated courier express dispatch hatch. Low-cost spatial demarcations, such as floor vinyl directional taping, high-visibility signage, and a dedicated staging wire shelf for completed digital orders, physically separate waiting drivers from dining room customers. This decoupling reduces frontline ambient chaos, eliminates driver queue friction, and restores operational line-of-sight for frontline cashiers.
3. **Dual-Check Poka-Yoke and Standardized Assembly Routines**: Human error during peak lunch order surges cannot be eliminated through verbal admonitions alone; it requires deterministic mistake-proofing mechanisms (Chase & Stewart, 1994). Restaurants must institutionalize the Two-Stage Fail-Safe Quality Verification Checklist (Table 16). Kitchen packers (Stage 1) must physically tick off meal components,

liquid condiment pouches, and cutlery against the printed order bill before sealing the bag. Frontline cashiers (Stage 2) must execute a mandatory visual and verbal verification routine, reading the four-digit GrabFood code aloud with the courier and confirming the order summary on the driver's smartphone before handover. Applying high-adhesion tamper-evident tape over box closures and bag handles provides visual integrity, prevents in-transit spills, and prevents courier tampering disputes.

4. **Dynamic Channel Inventory Synchronization and Stockout Governance**: Operational friction frequently arises when physical kitchen inventory de-synchronizes from the live digital storefront catalog. Selling depleted menu items forces cashiers into time-consuming phone negotiations with disappointed customers, increases driver waiting times, and risks platform cancellation penalties. Restaurant managers must implement dynamic inventory governance routines. Frontline supervisors must conduct a mandatory inventory audit at 11:00 WIB immediately prior to the lunch peak, toggling depleted or low-stock SKUs to "Sold Out" on the GrabFood Merchant portal. Additionally, promotional discount campaigns must be synchronized with daily ingredient procurement to ensure that discounted high-volume dishes do not stock out within the first hour of trading.

CONCLUSION

Online food delivery platforms offer substantial commercial growth for urban small restaurants, but long-term profitability and customer retention depend fundamentally on operational fulfillment rigour. Through an empirical single-case investigation of Nasi Gerilya on GrabFood in Medan City, this study demonstrates that high culinary quality cannot compensate for fulfillment failures. When platform customers pay a 23%–25% pricing mark-up, packaging omissions and dispatch errors trigger severe asymmetric rating penalties, eroding customer goodwill and generating substantial service recovery costs.

Our empirical analysis of 4,584 customer reviews, time-and-motion queue telemetry, and multi-informant fieldwork reveals that 64% of service failures stem from omitted condiments (38%) and courier waiting delays (26%). These defects originate not from kitchen culinary shortcomings, but from multi-channel queue crowding at an unsegregated frontline counter and the absence of standardized assembly mistake-proofing. To resolve these operational vulnerabilities, we proposed and validated three modular solutions: (1) a Two-Stage Fail-Safe Poka-Yoke packaging checklist, (2) a low-capital decoupled servicescape layout isolating delivery couriers, and (3) a Hierarchical Digital Service Recovery Framework with defined operational SLAs.

Limitations and Future Research

This study is bounded by its single-case design focused on an authentic Padang culinary establishment in Medan City operating on the GrabFood platform. While single-case studies provide deep, granular access to operational mechanisms, the findings may reflect specific cuisine characteristics, such as the heavy reliance on liquid sauces and separate condiment pouches. Future research should pursue multi-case comparative designs across diverse culinary formats (e.g., western fast food, beverage kiosks) and examine multi-platform fulfillment operations across GrabFood, GoFood, and ShopeeFood. Longitudinal empirical studies tracking defect rate reductions, net margin improvements, and customer repurchase intervals before and after implementing poka-yoke packaging protocols would further advance service operations literature in emerging digital economies.

ACKNOWLEDGMENTS

We thank the management and staff of Nasi Gerilya Restaurant Medan for their cooperation during fieldwork. We also thank the APINDO UMKM Merdeka program for supporting this research.

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